2. Fragments and Activity Interaction

4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Project Name: FragmentLifecycleDemo

Create a new fragment class:

java->new-> fragment->fragment(blank)

Fragment Name: DemoFragment

Fragment Layout name: fragment_demo

I

Filename: DemoFragment.java

```
package com.example.fragmentlifecycledemo;

import android.content.Context;

import android.os.Bundle;

import android.util.Log;

import android.view.LayoutInflater; import android.view.View;

import android.view.ViewGroup;

import androidx.annotation.NonNull; import androidx.annotation.Nullable;

import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

   String TAG = "FragmentLifecycle";


   @Override

   public void onAttach(@NonNull Context context) {

   super.onAttach(context);
```

```
   Log.d(TAG, "onAttach() executed");}



   @Override

  public void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       Log.d(TAG, "onCreate() executed");

   }



   @Override

   public View onCreateView(LayoutInflater inflater, ViewGroup
container,Bundle savedInstanceState) {

       Log.d(TAG, "onCreateView() executed");

       return inflater.inflate(R.layout.fragment_demo, container, false);

   }



@Override

public void onViewCreated(@NonNull View view, @Nullable Bundle
savedInstanceState) {

       super.onViewCreated(view, savedInstanceState);

        Log.d("on view created invoked", "onViewCreated() executed");

}



   @Override

   public void onStart() {

       super.onStart();

       Log.d("on start invoked", "on started");
```

```
}


@Override
public void onResume() {
    super.onResume();
    Log.d("on resume invoked", "on resume");
}


@Override
public void onPause() {
    super.onPause();
    Log.d("on pause invoked", "on pause");
}


@Override
public void onStop() {
    super.onStop();
    Log.d("on stop invoked", "on stopped");
}


@Override
public void onDestroyView() {
    super.onDestroyView();
    Log.d("on destroyview invoked", "on destroyview() executed");
}
```

```
    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }

    @Override
    public void onDetach() {
        super.onDetach();
        Log.d("on detach invoked", "on detach");
    }
}
```

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/fragment_container_view_tag"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello World!"

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toTopOf="parent" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: MainActivity.java

```
package com.example.fragmentlifecycledemo;


import android.os.Bundle;

import androidx.fragment.app.FragmentTransaction;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.graphics.Insets;

import androidx.core.view.ViewCompat;

import androidx.core.view.WindowInsetsCompat;


public class MainActivity extends AppCompatActivity {


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

                DemoFragment fragment = new DemoFragment();

                FragmentTransaction transaction =

                        getSupportFragmentManager().beginTransaction();

                transaction.replace(R.id.fragment_container_view_tag,
fragment);

                transaction.commit();

        }

    }
```

fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".DemoFragment">



    <!-- TODO: Update blank fragment layout -->

    <TextView

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:text="@string/hello_blank_fragment" />



</FrameLayout>
```